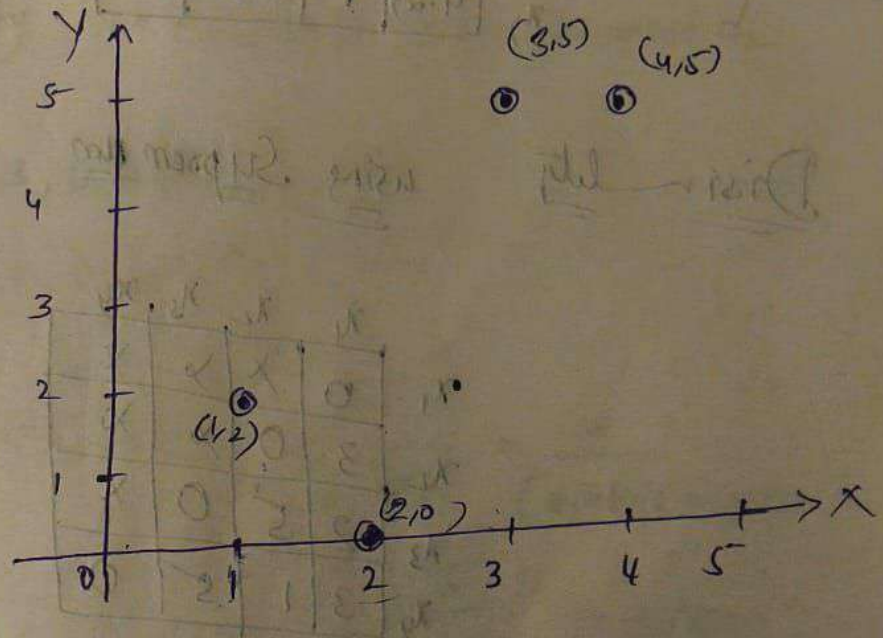


ACTIVITY - 5

Consider the following points

$(1,2)$ $(3,5)$ $(2,0)$ $(4,5)$. Construct the data matrix
& Similarity matrix using Mahattan, Euclidean,
Supremum measures.



Data matrix:-

Point	attribute 1	attribute 2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

Dissimilarity matrix using Manhattan measure

	x_1	x_2	x_3	x_4
x_1	0	x	x	x
x_2	5	0	x	x
x_3	3	6	0	x
x_4	6	1	7	0

$$d = |x_1 - x_2| + |x_3 - x_4|$$

Dissimilarity using Euclidean

	x_1	x_2	x_3	x_4
x_1	0	x	x	x
x_2	3.61	0	x	x
x_3	2.23	5.09	0	x
x_4	4.24	1	5.38	0

$$d = \sqrt{(x_1 - x_2)^2 + (x_3 - x_4)^2}$$

Dissimilarity using Supremum

	x_1	x_2	x_3	x_4
x_1	0	x	x	x
x_2	3	0	x	x
x_3	2	5	0	x
x_4	3	1	5	0

$$d = \max |x_i - x_j|$$

Finding the similarity measure b/w two documents using cosine similarities:

Document	team	coach	h	b	s	P	sc	W	L	sea
1	5	0	3	0	2	0	0	2	0	0
2	3	0	2	0	1	1	0	1	0	1
3	0	7	0	2	1	0	0	3	0	0
4	0	1	0	0	1	2	2	0	3	0

Cosine Similarity =

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \|d_2\|}$$

finding similarity b/w document d_1 & document d_2

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$\begin{aligned} \cos(d_1, d_2) &= \frac{25}{\sqrt{42} \cdot \sqrt{17}} = \frac{25}{(6.481)(4.12)} \\ &= \frac{25}{26.70} = 0.936 \end{aligned}$$